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Official Technical Roadmaps & Architecture Guides

HIGH-SCALE ARCHITECTURE

WhatsApp: 2 Billion Users with 50 Engineers

How WhatsApp serves 2 billion users and 100 billion messages daily using Erlang BEAM micro-processes and FreeBSD kernel optimizations.

1. Erlang BEAM Micro-Processes

Instead of heavy OS threads, each active user connection runs inside an ultra-light Erlang process consuming only ~2KB memory.

```
1 FreeBSD Server = 2,800,000 Concurrent TCP Sockets
```

2. Zero Database Bottleneck

WhatsApp does not store delivered messages on disk. Ephemeral message queues deliver packets and immediately purge from RAM.

- In-memory Mnesia & Redis queues
- Zero long-term message storage load

3. FreeBSD Kernel Tuning

Customized kernel network stack with enlarged file descriptor limits, zero-copy buffer allocations, and custom poll() event rings.

```
sysctl kern.ipc.maxsockets=4000000
sysctl kern.maxfiles=8000000
```

4. Lessons for Engineering Teams

Fewer moving parts = massive stability. Avoid complex distributed microservices where simple monolithic concurrency excels.

- 50 engineers vs 5000
- 99.999% uptime for 15+ years